

Degradation in NCM cathode materials

Problem

- High-Ni NCM provides high energy density.
- Higher Ni also accelerates degradation.
- Microstructural stability determines lifetime.

Knowledge gap

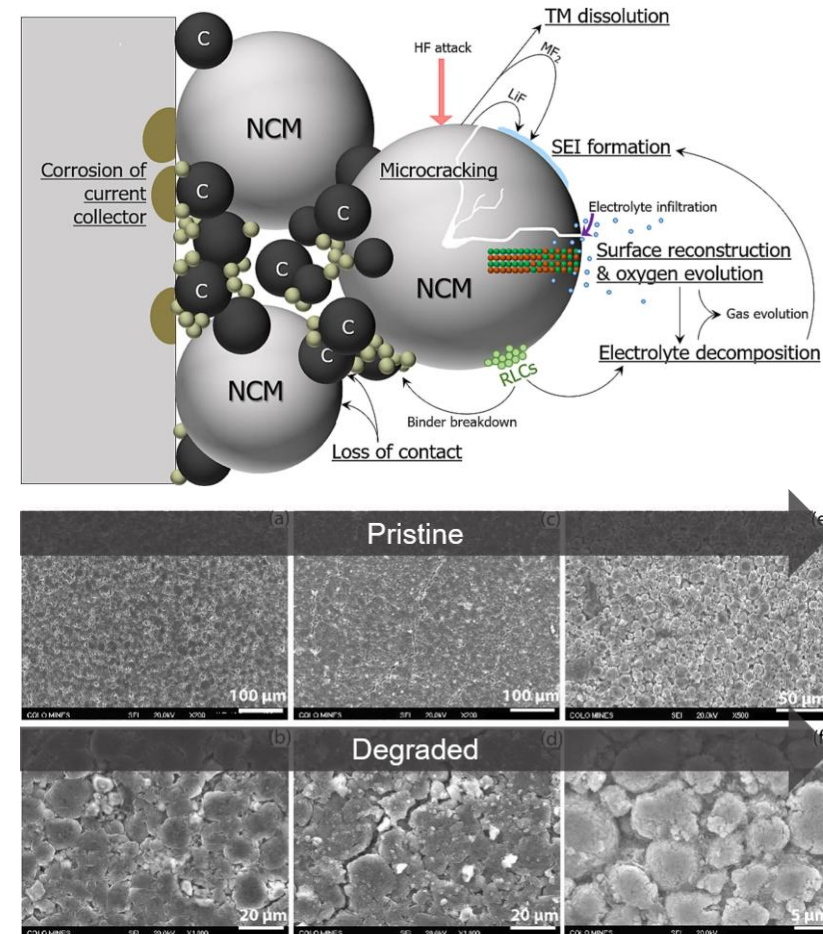
- Degradation is usually identified only after long cycling and post-mortem analysis.

Research question

- Can pristine SEM morphology predict future degradation behavior?

Implication

- Early screening of compositions and synthesis conditions
- Reduced experimental cost and testing time.
- Better materials-design decisions.



This work aims to transform SEM images from descriptive post-mortem observations into predictive, composition-aware microstructure signals for NCM cathode design.

Our previous work on convolutional neural networks (CNNs) for NCM degradation prediction

Our previous work

npj | computational materials

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Article



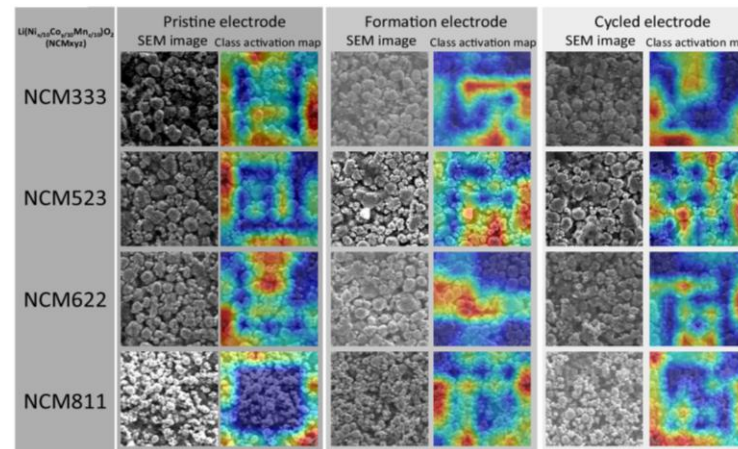
<https://doi.org/10.1038/s41524-024-01279-6>

Composition and state prediction of lithium-ion cathode via convolutional neural network trained on scanning electron microscopy images

Check for updates

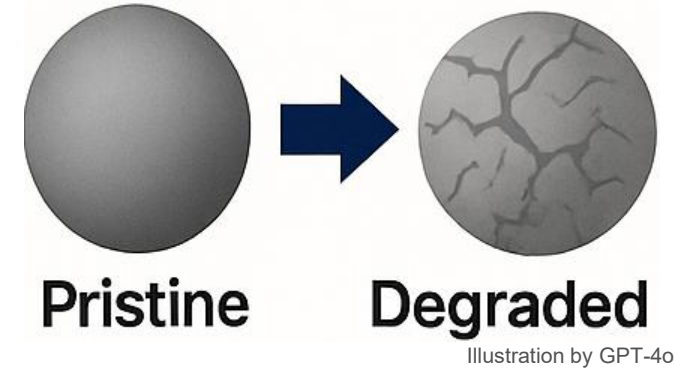
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The first two authors equally contributed



Current work

AC-GAN learns the mapping:



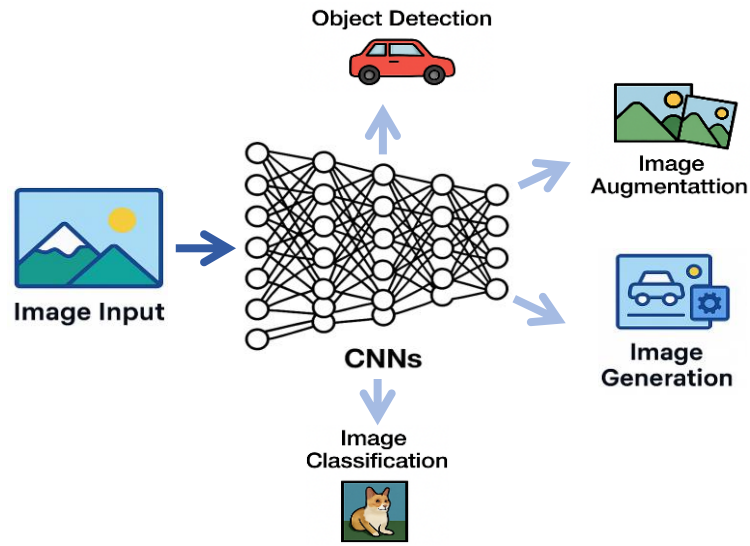
Pristine SEM + Composition → Degraded SEM

- Previous CNN work showed that SEM images contain information about NCM composition and degradation state.
- However, classification is retrospective → it only identifies degradation after it occurs.
- This work moves from classification to generation by predicting what degraded morphology may look like from pristine SEM images.

Materials characterization via convolutional neural networks (CNNs)

- CNNs

Employ a neural network architecture designed to process image data



$$y_{i,j,k} = \sum_m \sum_n x_{i+m,j+n} * w_{m,n,k} + b_k$$

$y_{i,j,k}$: the output at the position (i,j) for the k^{th} filter

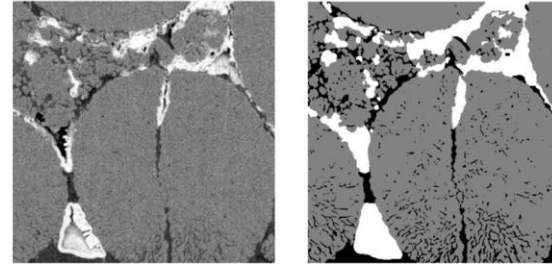
$x_{i+m,j+n}$: the input value at the position $(i + m, j + n)$

$w_{m,n,k}$: represents the filter weights

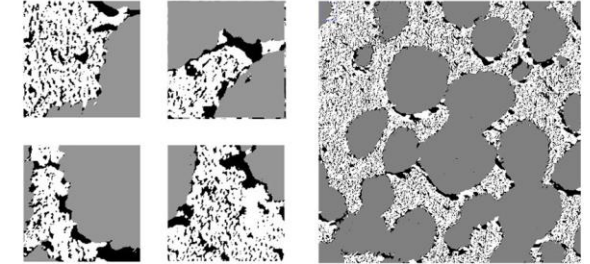
b_k : the bias term

- CNN for Materials Characterization

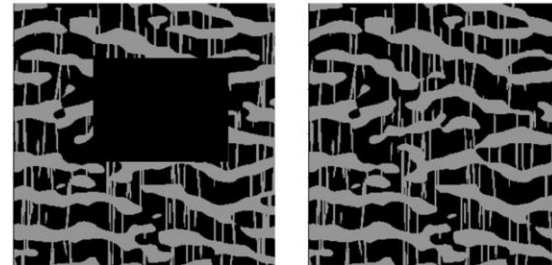
Segmentation



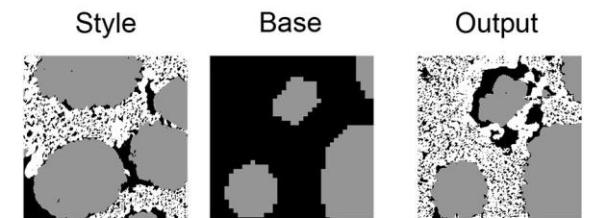
Generation



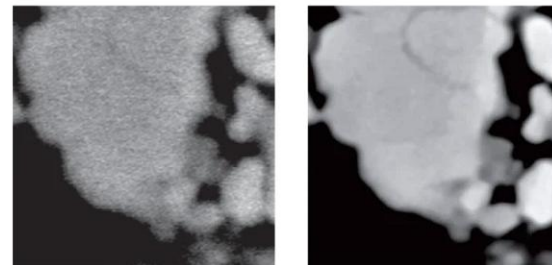
Inpainting



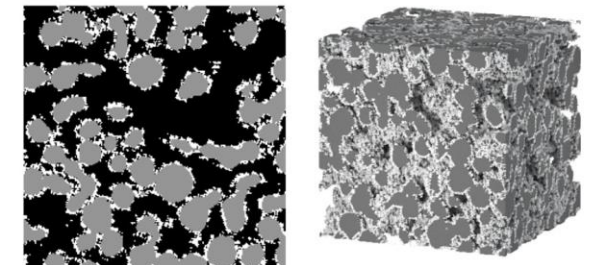
Style transfer



Super-resolution



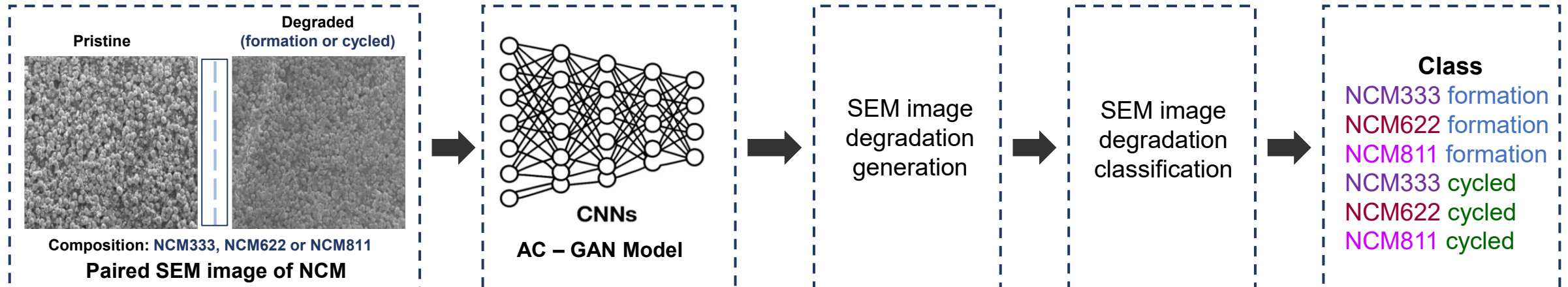
Dimensionality expansion



Goal

Goal 1:

To generate **degraded** NCM morphology (**formation** or **cycled**) from **pristine** NCM morphology **conditioned** on composition (**NCM333**, **NCM622**, or **NCM811**).

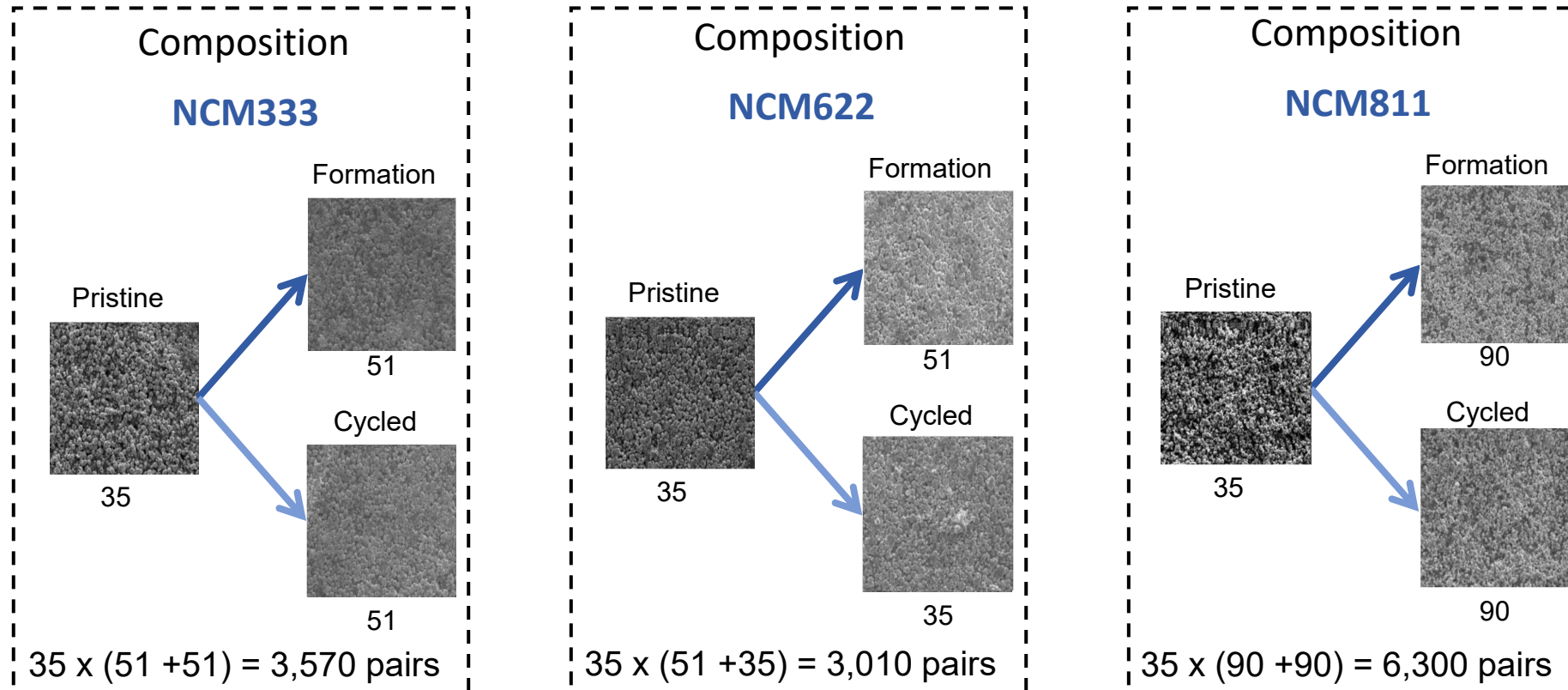


Formation: SEM image after the formation cycle
Cycled: SEM image after 100 cycles

Instead of **only** telling us **what degradation** has already occurred, the model predicts what degradation will look like **visually** in the future if we know the pristine image and give it to the AC-GAN model to learn from.

Dataset construction

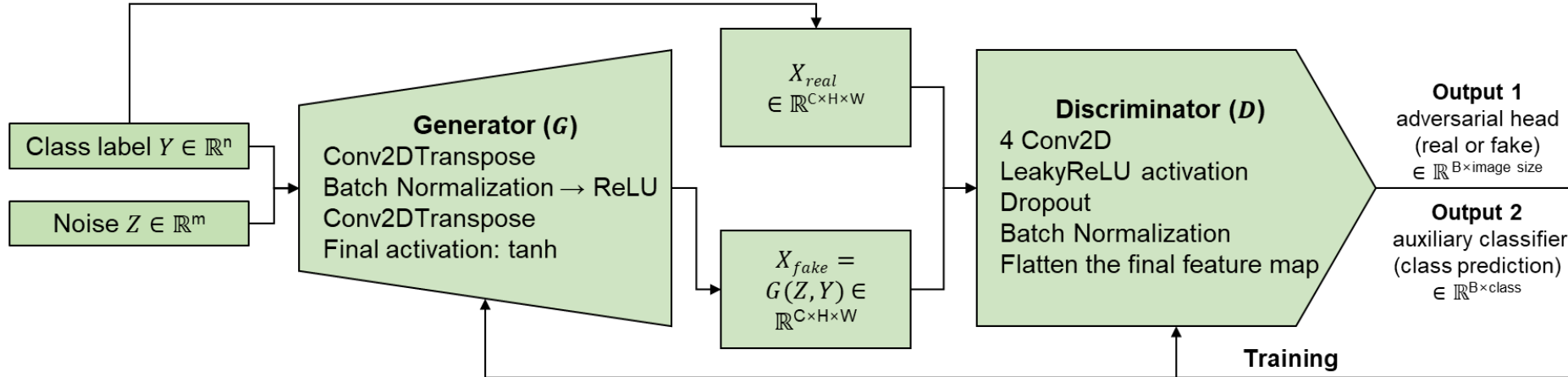
- Dataset construction
 - Max. 35 images per pristine and Max. 90 images per degraded (formation and cycled)



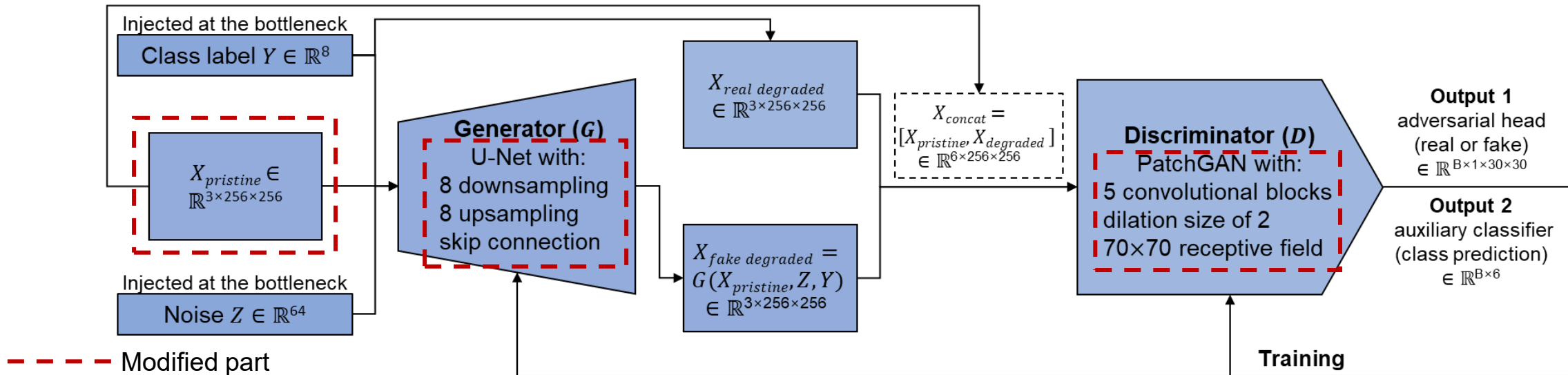
Total paired pristine–degraded NCM image constructed: 12,880 images

AC – GAN Model

- Standard AC-GAN model architecture



- Our modified AC-GAN Model architecture (used in this work)



Loss Function

- Adversarial loss: hinge loss for real/fake patch maps

$$D: \mathcal{L}_{D \text{ adv}} = E_{\text{real}}[\max(0, 1 - D_{\text{adv}}^{\text{real}})] + E_{\text{fake}}[\max(0, 1 + D_{\text{adv}}^{\text{fake}})]$$

$$G: \mathcal{L}_{G \text{ adv}} = -E_{\text{fake}}[D_{\text{adv}}^{\text{fake}}]$$

- Auxiliary classification loss: Cross-Entropy (CE) loss for both real and fake

$$\mathcal{L}_{\text{aux}} = E_{\text{real}}[CE(D_{\text{class}}^{\text{real}}, Y)] + E_{\text{fake}}[CE(D_{\text{class}}^{\text{fake}}, Y)]$$

- Generator extra losses

$$\text{L1 Loss: } \mathcal{L}_{L1} = \|X_{\text{degraded}} - \hat{X}_{\text{degraded}}\|_1$$

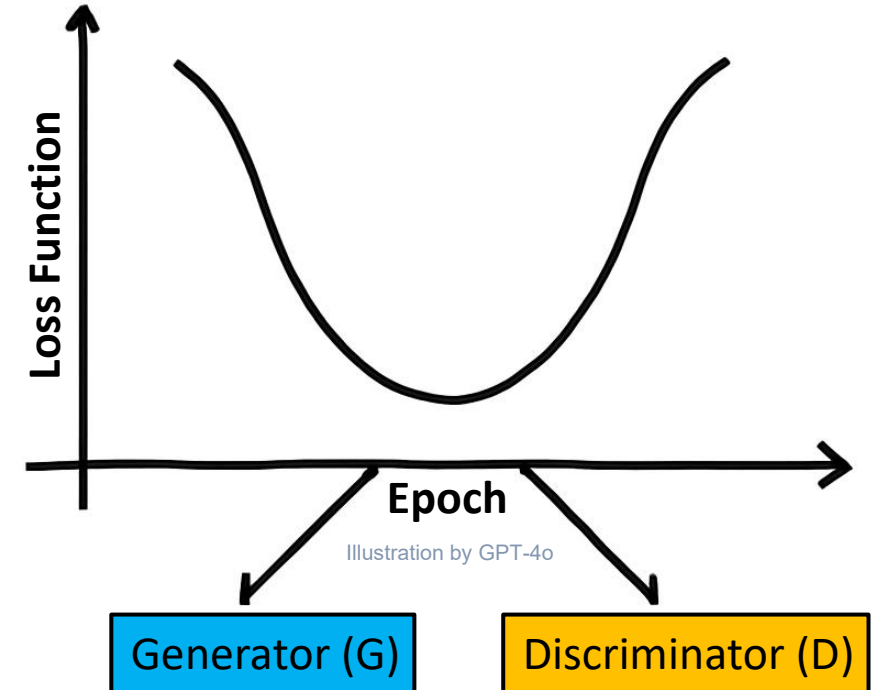
$$\text{VGG Perceptual Loss: } \mathcal{L}_{\text{perc}} = \|\phi(X_{\text{degraded}}) - \phi(\hat{X}_{\text{degraded}})\|_1$$

$$\text{Feature Matching (FM) Loss: } \mathcal{L}_{\text{FM}} = \|D_{\text{features}}(X_{\text{real}}) - D_{\text{features}}(\hat{X}_{\text{degraded}})\|_1$$

- Final combined loss

$$\text{D loss total : } \mathcal{L}_D = \mathcal{L}_{D \text{ adv}} + \lambda_{\text{aux}} \cdot \mathcal{L}_{\text{aux}}$$

$$\text{G loss total : } \mathcal{L}_G = \lambda_{\text{adv}} \cdot \mathcal{L}_{G \text{ adv}} + \lambda_{\text{aux}} \cdot \mathcal{L}_{\text{aux}} + \lambda_{L1} \cdot \mathcal{L}_{L1} + \lambda_{\text{perc}} \cdot \mathcal{L}_{\text{perc}} + \lambda_{\text{FM}} \cdot \mathcal{L}_{\text{FM}}$$



$$\begin{array}{ll} \lambda_{\text{adv}} = 5.0 & \lambda_{L1} = 2.0 \\ \lambda_{\text{aux}} = 1.5 & \lambda_{\text{perc}} = 0.05 \\ \lambda_{\text{FM}} = 2.0 & \end{array}$$

Why DISTS is appropriate for **generated (fake)** NCM SEM degradation image evaluation

Microstructure image problem

- SEM degradation is not pixel-perfect.
- Small shifts or different particle positions are acceptable if key morphology features are preserved, such as roughening, cracks, pores, voids, and surface texture.

Common metrics are weak

- Common metrics (PSNR, MSE, FID, and KID) are limited and pixel-based
- PSNR and MSE are too sensitive to small pixel shifts.
- MS-SSIM emphasizes structural/luminance similarity, but may miss fine degradation.
- FID and KID measure image realism, but not how well each generated SEM image matches its paired reference.

Why DISTS fits this task

- DISTS is well-suited for this task because it measures both structural and textural similarity.
- It is less sensitive to small geometric changes than pixel-based metrics.
- For SEM images, preserving degradation features such as cracks, pores, and surface texture is more important than matching exact pixel values.

PSNR: Peak Signal-to-Noise Ratio

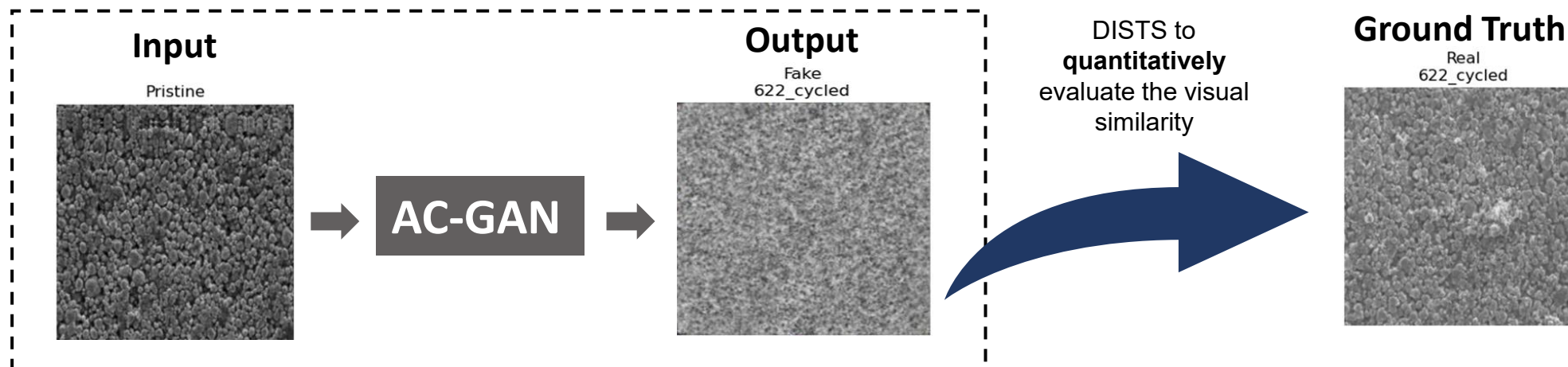
MS-SSIM: Multi-Scale Structural Similarity Index Measure

FID: Fréchet Inception Distance

KID: Kernel Inception Distance

DISTS: Deep Image Structure and Texture Similarity

MSE: mean squared error



AC – GAN training

• Training configuration

- Optimized with Adam optimizer:
 - G: learning rate = 1e-4
 - D: learning rate = 1.1e-4
- Learning rate scheduling: Cosine Annealing
- Batch size: 32

• Metrics Performance

- Fidelity and realism : FID
- Similarity to ground truth: DISTS
- Semantic consistency: auxiliary classification accuracy

• Composite score

Best trained model selection using a composite function of **FID** z-score and **auxiliary classification accuracy** z-score.

$$Composite\ score_i = \lambda_{FID} \cdot \left(-\frac{FID_i - \mu_{FID}}{\sigma_{FID}} \right) + \lambda_{aux\ accuracy} \cdot \left(\frac{Aux\ accuracy_i - \mu_{aux\ accuracy}}{\sigma_{aux\ accuracy}} \right) \uparrow \text{the higher, the better}$$

$$\lambda_{FID} = 0.9 \quad \lambda_{aux\ accuracy} = 0.1$$

μ_{FID} and $\mu_{aux\ accuracy}$ are the historical means for FID and auxiliary classification accuracy

σ_{FID} and $\sigma_{aux\ accuracy}$ are the historical standard deviations for FID and auxiliary classification accuracy

Loss Function

- Adversarial loss: hinge loss for real/fake patch maps

$$D: \mathcal{L}_{D \text{ adv}} = E_{\text{real}}[\max(0, 1 - D_{\text{adv}}^{\text{real}})] + E_{\text{fake}}[\max(0, 1 + D_{\text{adv}}^{\text{fake}})]$$

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- Generator extra losses

$$\text{L1 Loss: } \mathcal{L}_{L1} = \|X_{\text{degraded}} - \hat{X}_{\text{degraded}}\|_1$$

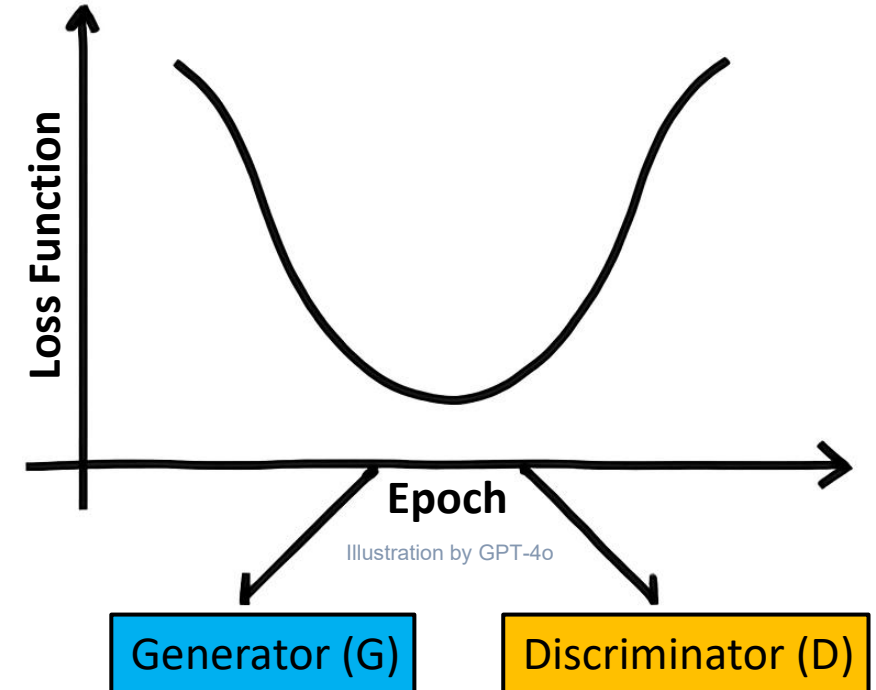
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$$\text{Feature Matching (FM) Loss: } \mathcal{L}_{\text{FM}} = \|D_{\text{features}}(X_{\text{real}}) - D_{\text{features}}(\hat{X}_{\text{degraded}})\|_1$$

- Final combined loss

$$\text{D loss total : } \mathcal{L}_D = \mathcal{L}_{D \text{ adv}} + \lambda_{\text{aux}} \cdot \mathcal{L}_{\text{aux}}$$

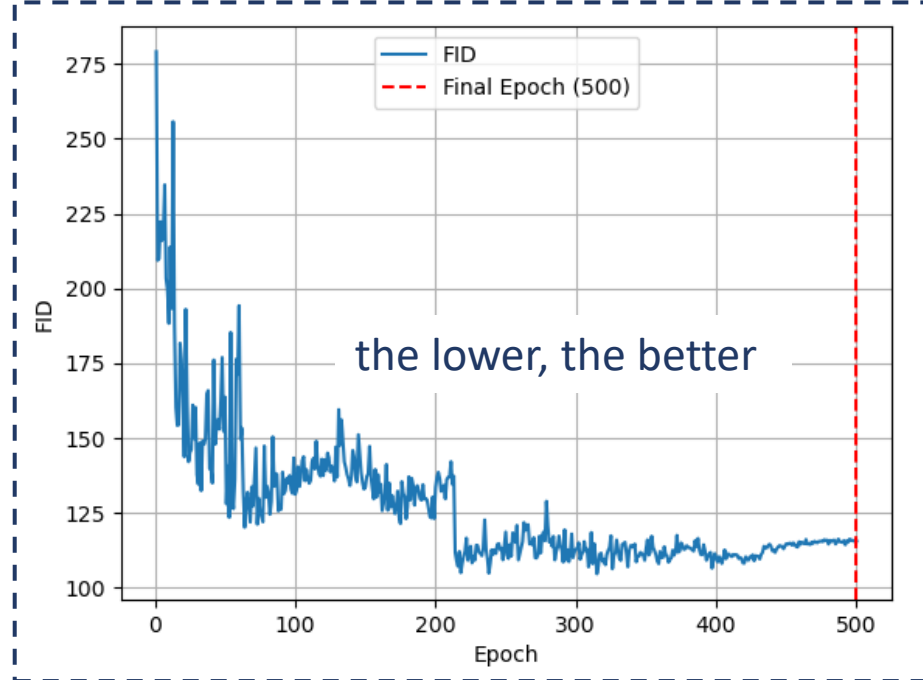
$$\text{G loss total : } \mathcal{L}_G = \lambda_{\text{adv}} \cdot \mathcal{L}_{G \text{ adv}} + \lambda_{\text{aux}} \cdot \mathcal{L}_{\text{aux}} + \lambda_{L1} \cdot \mathcal{L}_{L1} + \lambda_{\text{perc}} \cdot \mathcal{L}_{\text{perc}} + \lambda_{\text{FM}} \cdot \mathcal{L}_{\text{FM}}$$



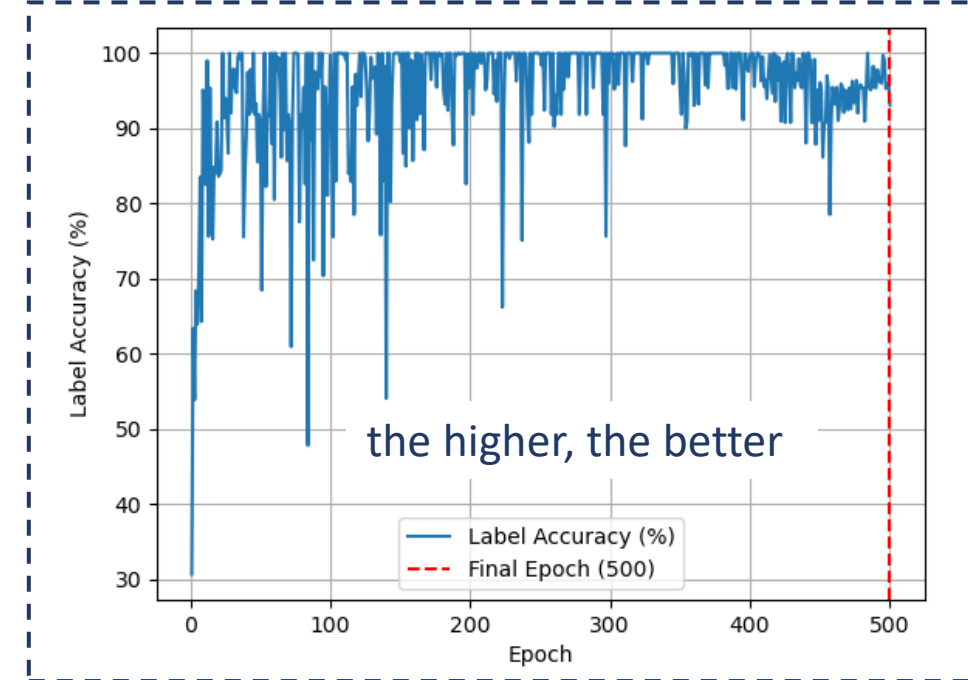
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Best trained model selection

FID value during training



Auxiliary (label) accuracy during training



$$Composite\ score_i = \lambda_{FID} \cdot \left(- \frac{FID_i - \mu_{FID}}{\sigma_{FID}} \right) + \lambda_{aux\ accuracy} \cdot \left(\frac{Aux\ accuracy_i - \mu_{aux\ accuracy}}{\sigma_{aux\ accuracy}} \right)$$

- The highest composite score at epoch 216 with value 7.33
- The trained model saved at epoch 216 is the best model

Class label classification for **real** and **generated** degradation

Auxiliary classification test accuracy of the **real** degraded SEM image

	Precision	Recall	f1-score	support
333_cycled	1.0000	0.4726	0.6419	146
333_formation	0.7273	1.0000	0.8421	152
622_cycled	1.0000	0.9744	0.9870	156
622_formation	0.9588	1.0000	0.9789	93
811_cycled	0.9867	0.8043	0.8862	276
811_formation	0.7898	0.9893	0.8784	281
Accuracy			0.8750	1104
Macro avg	0.9104	0.8734	0.8691	1104
Weighted avg	0.9021	0.8750	0.8679	1104

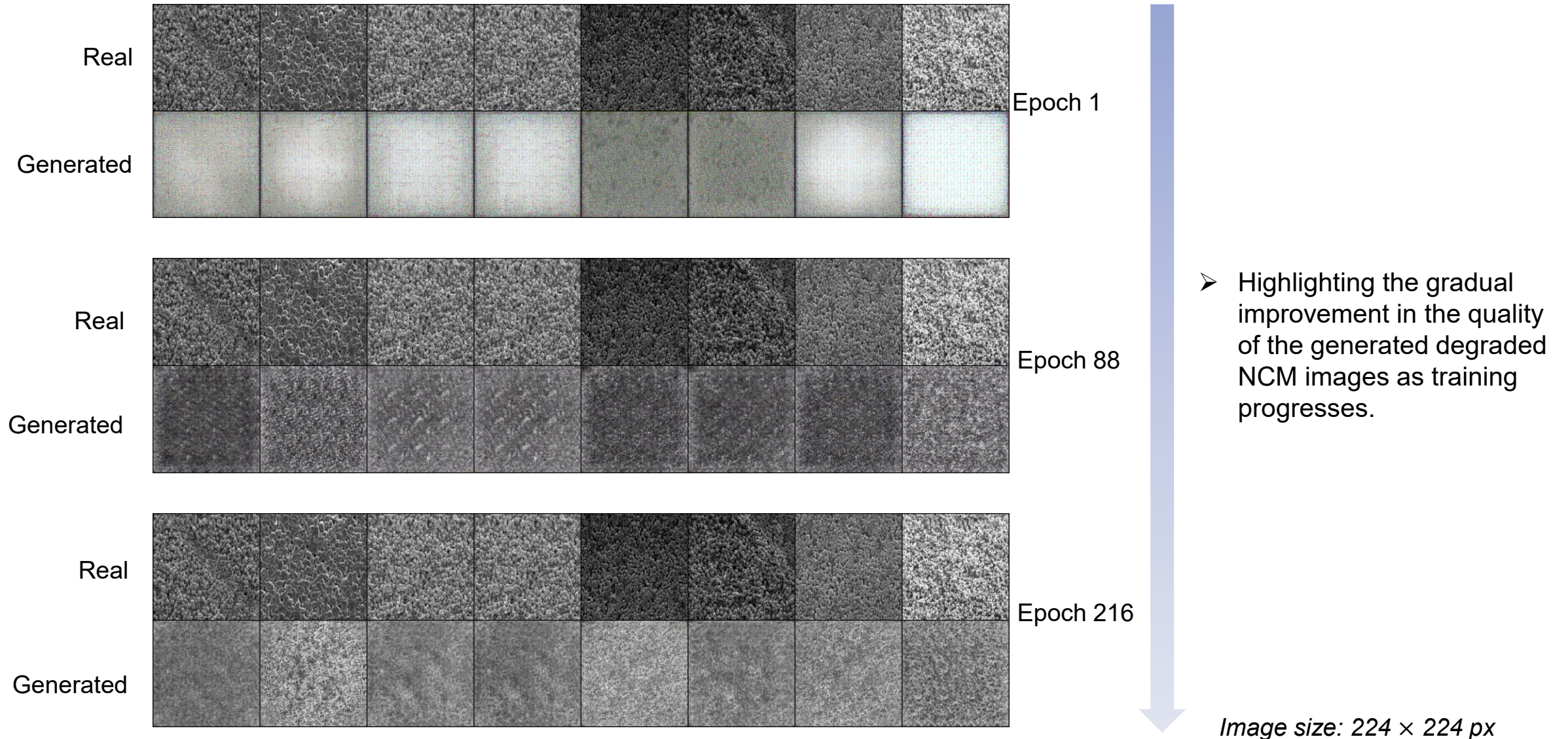
Accuracy: 87.50%

Classification test accuracy of the **generated** degraded SEM images

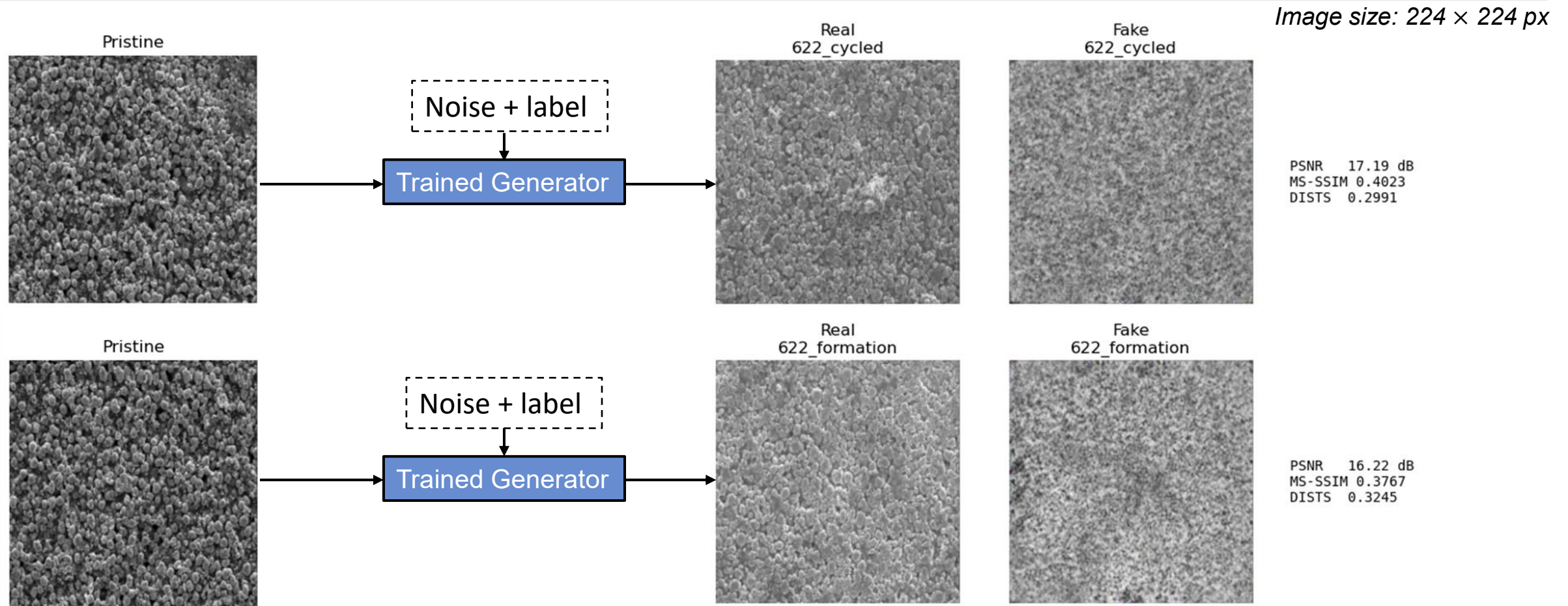
	Precision	Recall	f1-score	support
333_cycled	1.0000	1.0000	1.0000	146
333_formation	1.0000	1.0000	1.0000	152
622_cycled	1.0000	1.0000	1.0000	156
622_formation	1.0000	1.0000	1.0000	93
811_cycled	1.0000	0.9710	0.9860	276
811_formation	0.9723	1.0000	0.9860	281
Accuracy			0.9928	1104
Macro avg	0.9954	0.9952	0.9952	1104
Weighted avg	0.9930	0.9928	0.9928	1104

Accuracy: 99.28%

Generated degraded NCM Image evolution during training

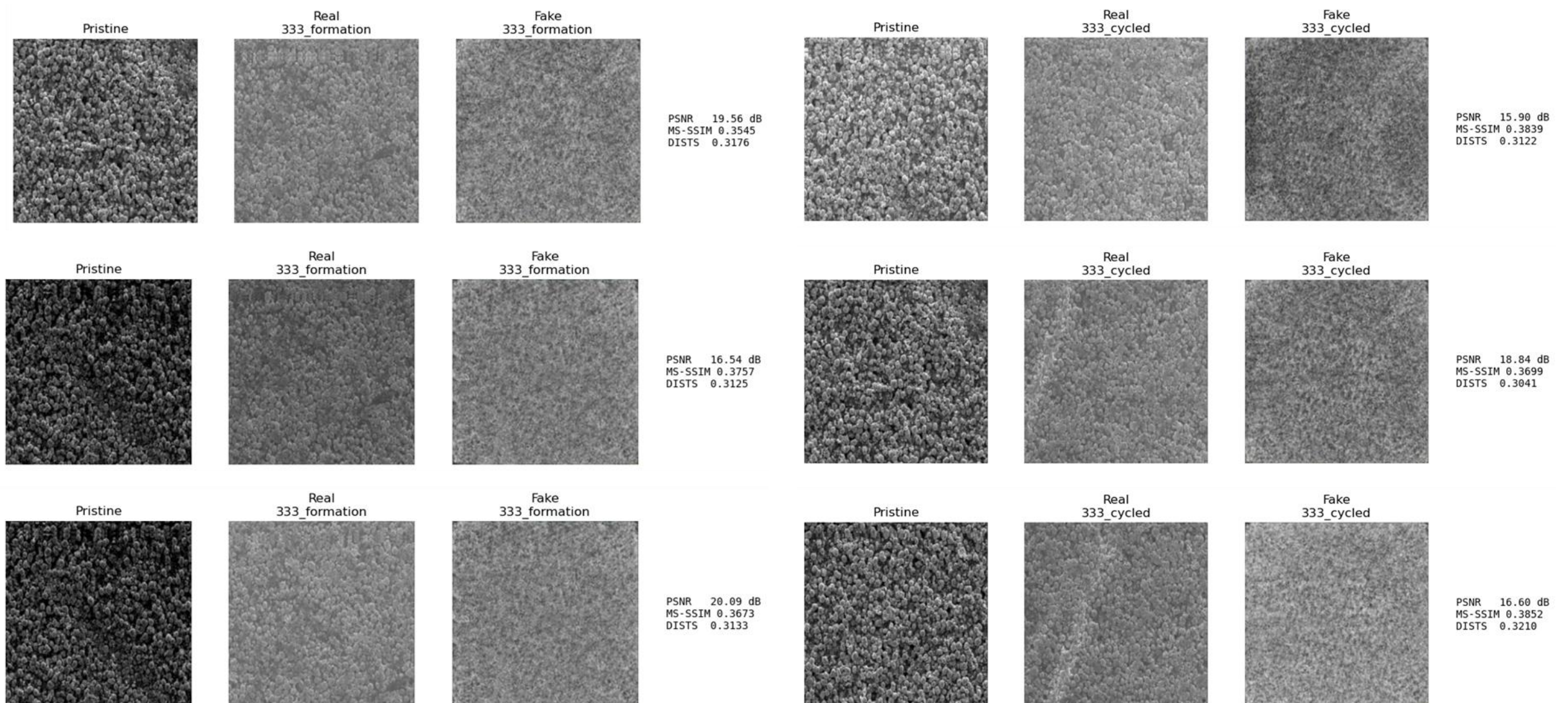


Samples of degraded NCM333 image generation test

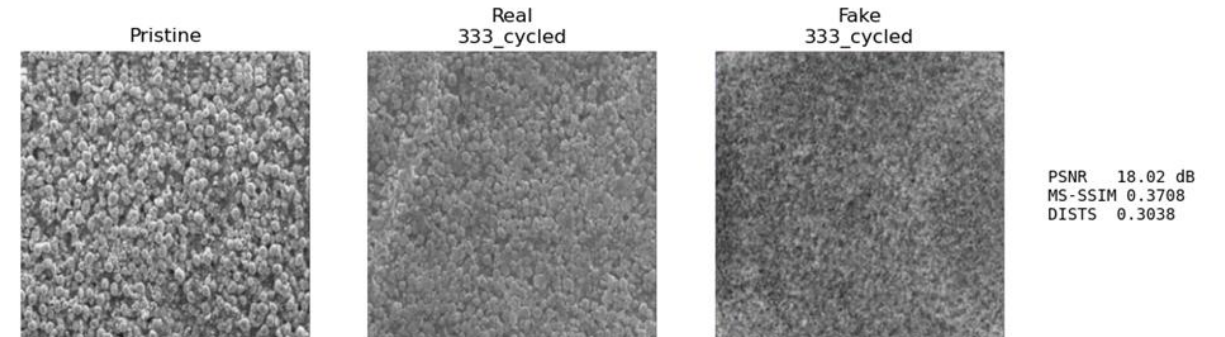
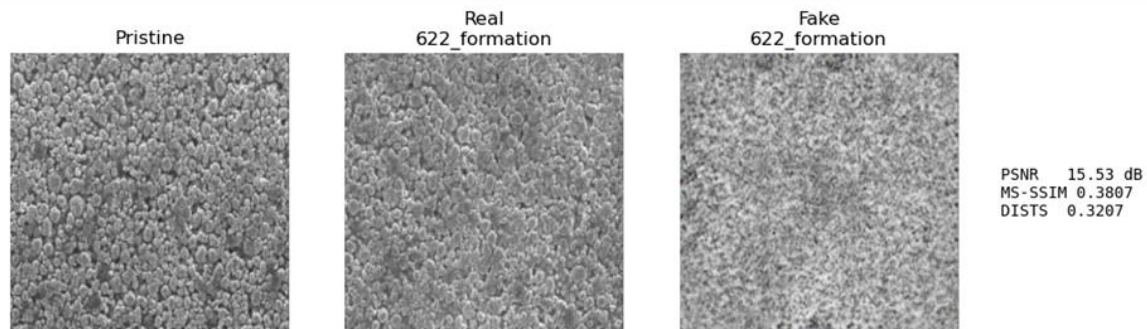
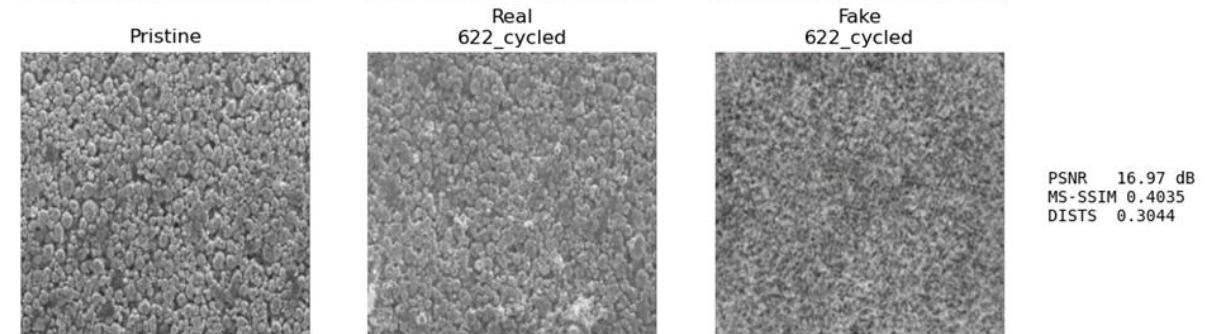
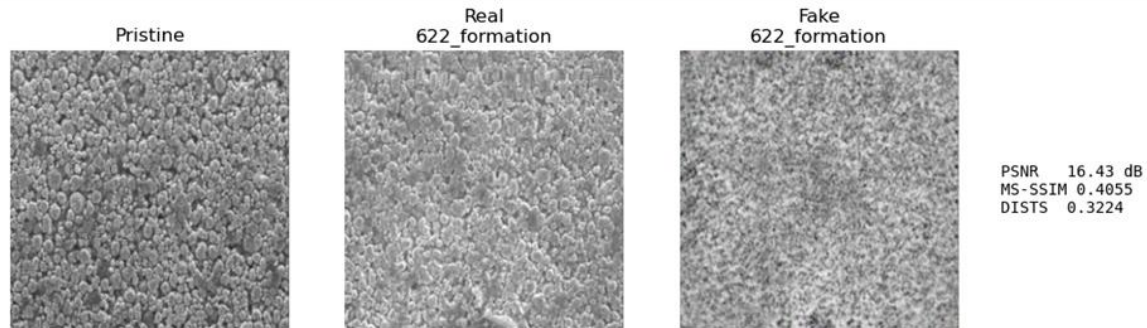
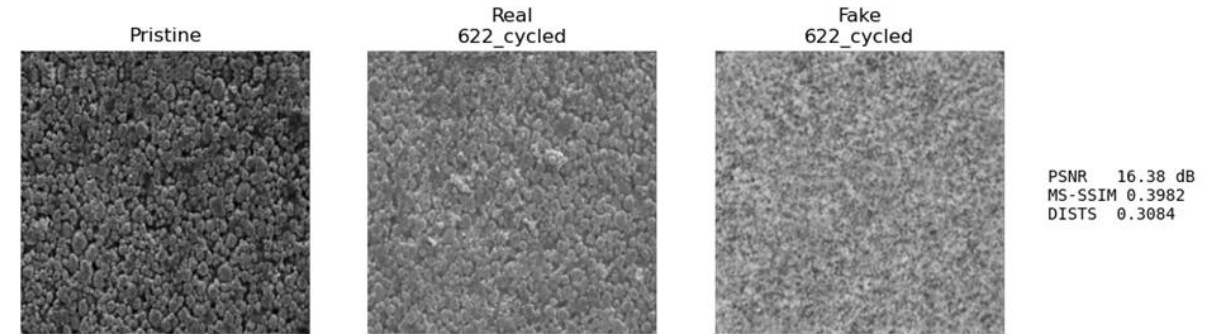
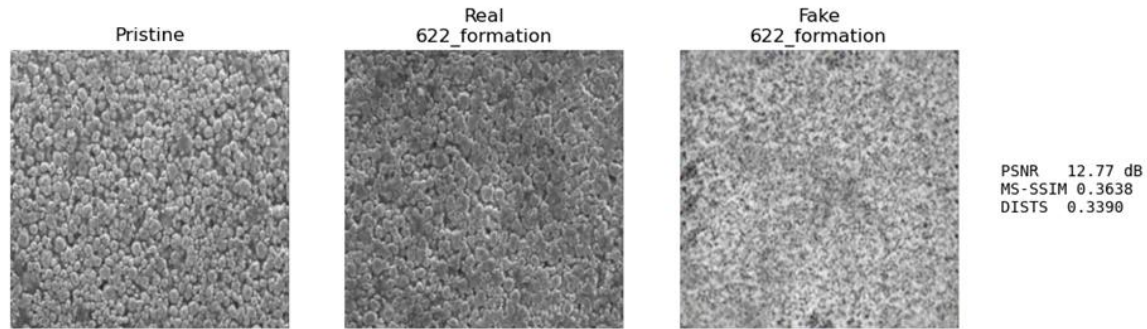


- Generated NCM333 images reproduce particle texture and degradation-like contrast. Low PSNR/MS-SSIM reflects pixel-level mismatch, while DISTS better captures morphology-level similarity
- This proves that the DISTS metrics are better than pixel-based metrics (PSNR and MS-SSIM)

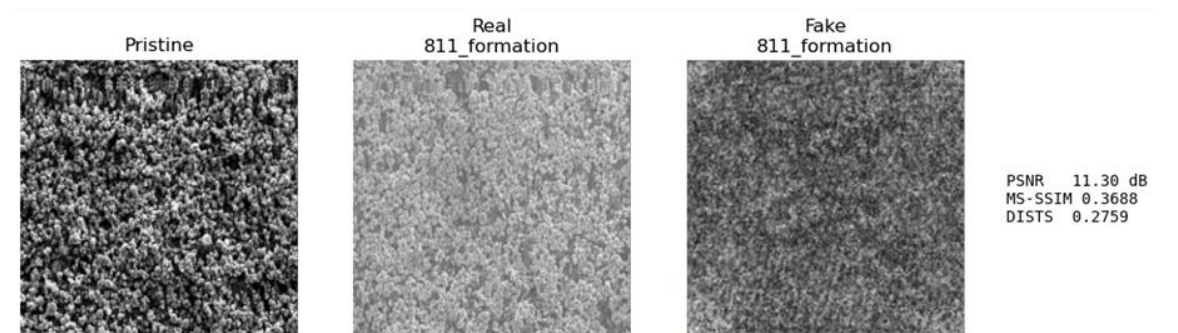
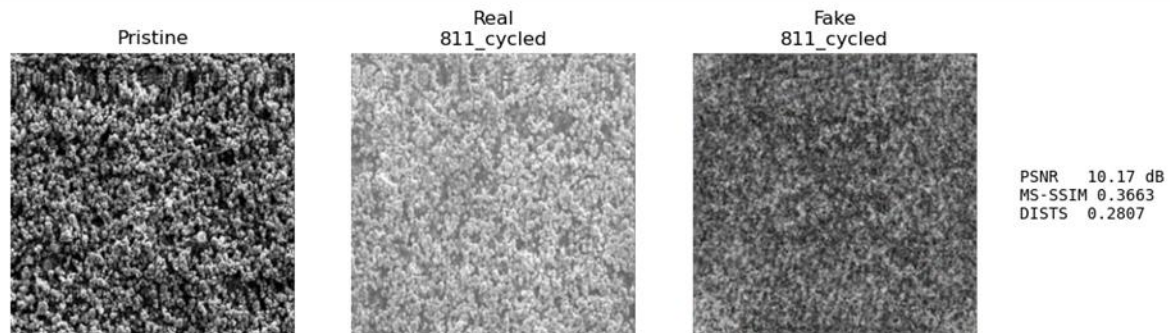
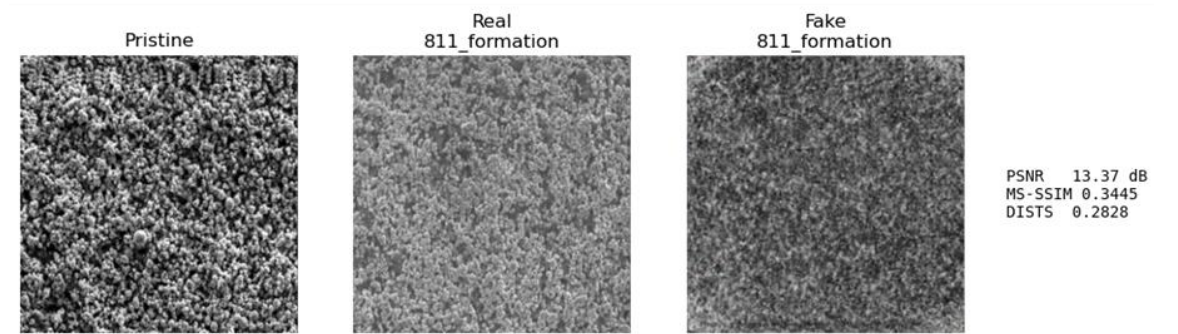
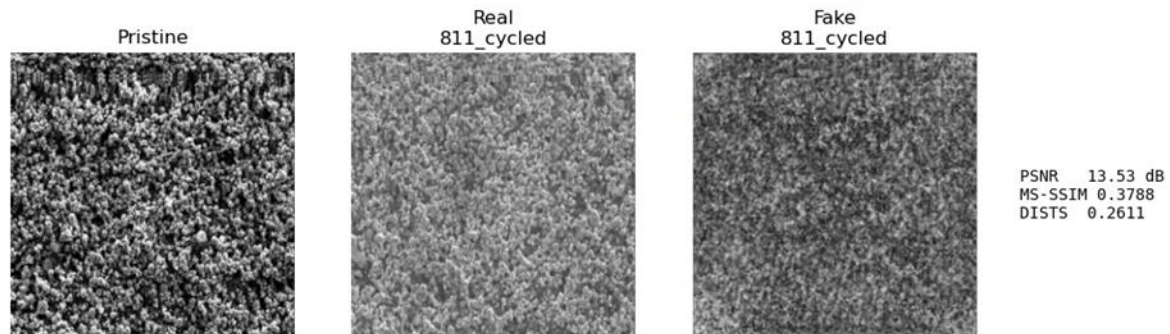
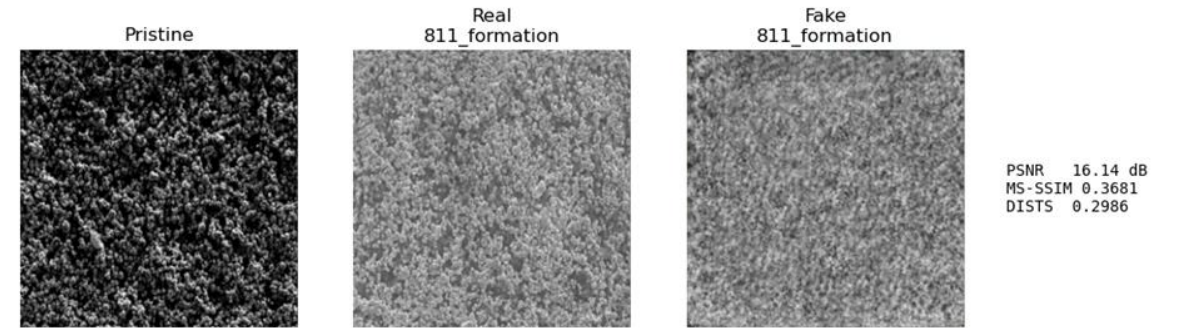
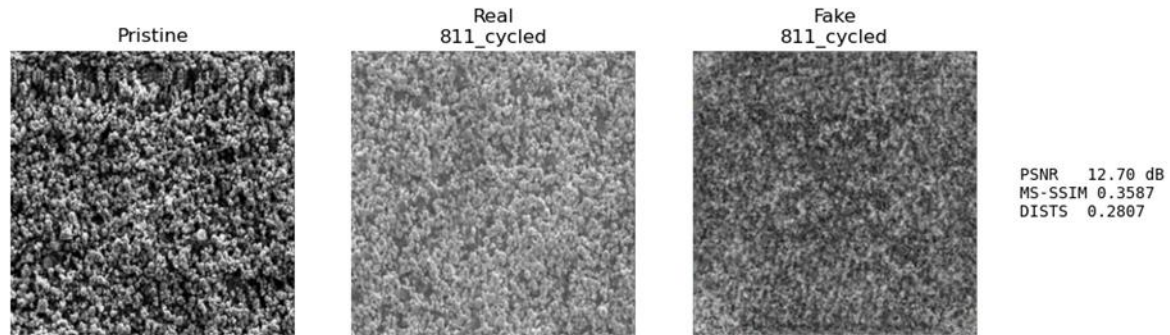
Other Generated Fake Degraded of NCM333 Images



Generated Fake Degraded of NCM622 Images



Generated Degraded of NCM811 Images



Summary

Conclusion:

- This work developed a class-conditioned AC-GAN framework to generate and classify SEM-based degradation morphology conditioned on composition.
- The generated images exhibited strong semantic consistency with the target degradation classes, achieving 99.28% auxiliary classification accuracy.
- Compared with PSNR and MS-SSIM, DISTSS is more suitable because it captures both structural and textural similarity rather than exact pixel-level agreement. This better reflects SEM degradation morphology and supports the generated images as plausible degradation-state hypotheses.
- These results demonstrate the potential of generative SEM modeling as a pre-screening tool for composition-aware degradation assessment, enabling earlier evaluation of NCM cathode materials before long-term cycling experiments.
- The current framework does not provide quantitative estimates of the proportions of individual degradation features (e.g., cracks, pores, and other microstructural defects) contributing to the overall degradation behavior.

Future Work:

- Will test the model on the out-of-distribution sample, such as SEM images of NCM materials containing additives.
- Will compare with the diffusion model to make the classification and generation.
- Will enhance the work impact by including quantification of the degradation features.